

Equation of a line

Basic skills

Q1. $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

a) $\lambda = 2 \quad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix} + 2 \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 9 \\ 8 \end{pmatrix}$

b) $\lambda = -2 \quad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix} - 2 \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$

c) $x = 5 + 2\lambda, \quad y = 2 + 3\lambda$

d) $x = 20 \quad 20 = 5 + 2\lambda \quad \lambda = \frac{15}{2} = 7.5$

so $y = 2 + 3 \times 7.5 = 24.5$

Q2. $A = (3, 4) \quad B = (5, -1)$

a) $\underline{d} = \underline{b} - \underline{a} = \begin{pmatrix} 5 \\ -1 \end{pmatrix} - \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$

could have any multiple of this including $\begin{pmatrix} -2 \\ 5 \end{pmatrix}, \begin{pmatrix} -4 \\ 10 \end{pmatrix}, \dots$

$$b) \quad \underline{r} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

↑ could replace with any point on line e.g. $\begin{pmatrix} 5 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 9 \end{pmatrix}, \dots$

$$c) \quad \begin{pmatrix} 2 \\ -5 \end{pmatrix} \quad \begin{array}{c} 2 \\ \text{vector of } \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} \\ -5 \\ m = \frac{\Delta y}{\Delta x} \end{array}$$

Competency skills

$$Q4. \quad A = (-1, 2, 5), \quad B = (6, 4, -3)$$

$$a) \quad \underline{d} = \underline{b} - \underline{a} = \begin{pmatrix} 6 \\ 4 \\ -3 \end{pmatrix} - \begin{pmatrix} -1 \\ 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 7 \\ 2 \\ -8 \end{pmatrix}$$

$$\underline{r} = \begin{pmatrix} -1 \\ 2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 2 \\ -8 \end{pmatrix}$$

$$b) \quad \underline{r} = \begin{pmatrix} 6 \\ 4 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 2 \\ -8 \end{pmatrix} \text{ or } \begin{pmatrix} -1 \\ 2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 14 \\ 4 \\ -16 \end{pmatrix}$$

c) $x = 7\lambda - 1, y = 2\lambda + 2, z = 5 - 8\lambda$

$$\lambda = \frac{x+1}{7}, \quad \lambda = \frac{y-2}{2}, \quad \lambda = \frac{z-5}{-8}$$

so $\frac{x+1}{7} = \frac{y-2}{2} = \frac{z-5}{-8}$

Q5. $\frac{x-5}{2} = \frac{y-2}{-3} = \frac{z+1}{5}$

so $\underline{a} = \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix} \quad \underline{d} = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$

so $\underline{r} = \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$

$\lambda = \lambda$
can
get

$x = 5 + 2\lambda$

$y = 2 - 3\lambda$

$z = -1 + 5\lambda$

Q6. $\underline{r} = \lambda (2\underline{i} - \underline{j} + 4\underline{k})$

so $x = 2\lambda, y = -\lambda, z = 4\lambda$

$$\frac{x}{2} = -y = \frac{z}{4}$$

Q7. $x = 2, y = 2z$

so $\lambda = y, \lambda = 2z$

$x = 2 + 0\lambda$ so $y = \lambda, z = \frac{\lambda}{2}$

$$\underline{r} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ \frac{1}{2} \end{pmatrix}$$

Q8. Let $\underline{r} = \begin{pmatrix} 10 \\ -3 \\ -1 \end{pmatrix}$

so $1 - 3\lambda = 10$ so $\lambda = -3$
 $\lambda = -3$ solves all
 $2 - \lambda = -1$

so yes it does (but won't go through $(10, -4, -1)$)



Advanced skills

Q9. a) $(3, -2, 1)$ and $(5, 0, 2)$

$$\underline{d} = \begin{pmatrix} 5 \\ 0 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$$

$$\Gamma_1 = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$$

b) $3x - 6 = 2y + 1 = z - 5$

$$\frac{x-2}{\frac{1}{3}} = \frac{y+\frac{1}{2}}{\frac{1}{2}} = \frac{z-5}{1}$$

so $\Gamma_2 = \begin{pmatrix} 2 \\ -\frac{1}{2} \\ 5 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$ ← $\begin{pmatrix} \frac{1}{3} \\ \frac{1}{2} \\ 1 \end{pmatrix}$ Scaled up

diff parameter

c) As $\begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \neq k \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$ for any k

the lines are not parallel

d) using $\cos \theta = \frac{a \cdot b}{|a||b|}$

$$\cos \theta = \frac{2 \times 2 + 2 \times 3 + 1 \times 6}{\sqrt{2^2 + 2^2 + 1^2} \sqrt{2^2 + 3^2 + 6^2}} = \frac{16}{21}$$

$$\theta = 40.4^\circ$$

e) solving $\underline{r}_1 = \underline{r}_2$

$$\begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -\frac{1}{2} \\ 5 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$$

$$\begin{array}{l} \text{so} \quad \left. \begin{array}{l} 3 + 2\lambda = 2 + 2\mu \\ -2 + 2\lambda = -\frac{1}{2} + 3\mu \\ 1 + \lambda = 5 + 6\mu \end{array} \right\} \begin{array}{l} \text{check} \\ \text{solving} \end{array} \end{array}$$

$$\text{so } 2\mu = 2\lambda + 1 \quad \mu = \lambda + \frac{1}{2}$$

$$\text{then } -2 + 2\lambda = -\frac{1}{2} + 3\lambda + \frac{3}{2}$$

$$\text{so } \lambda = -3, \mu = -\frac{5}{2}$$

now check final

$$1 + -3 = 5 + 6 \times -\frac{5}{2}$$

$$-2 = -10 \quad \times \quad \text{NO}$$

so they do not intersect

Q10. $\underline{r} = 3t\underline{i} + t^2\underline{j} + (e^t - 2)\underline{k}$

at $t=0$ $\underline{r} = -\underline{k}$

at $t=10$ $\underline{r} = 30\underline{i} + 100\underline{j} + (e^{10} - 2)\underline{k}$

so vector between is

$$\begin{pmatrix} 30 \\ 100 \\ e^{10} - 1 \end{pmatrix}$$

magnitude

$$\sqrt{30^2 + 100^2 + (e^{10} - 1)^2}$$

$$= 22025$$